

Brain Tumor MRI Classification Using Convolutional Neural Networks and Transfer Learning

6CS012 Final Portfolio Project | Herald College Kathmandu | Part II — Vision Task

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GitHub: github.com/prashunbaral/AIML — Assessment/CNN.ipynb

Abstract

Objective. This project addresses automated brain tumor classification from MRI scans — a task where classification speed and accuracy directly shape clinical decisions.

Methods. Four deep learning models are trained and compared on Kaggle's Brain Tumor MRI Dataset (7,023 labelled images; four classes: glioma, meningioma, no tumor, pituitary). Part A covers three custom CNN architectures built from scratch: a lightweight three-block baseline, and a five-block VGG-style deeper network trained with both Adam and SGD, incorporating Batch Normalization, SpatialDropout, GlobalAveragePooling, and L2 regularization. Part B applies transfer learning via ResNet50 (ImageNet pretrained) using a two-phase protocol: feature extraction with the backbone frozen, followed by fine-tuning the top 30 layers at a lower rate.

Key Findings. ResNet50 achieves ~96% test accuracy and 0.99 ROC-AUC in just 20 epochs, outperforming all scratch models that ran up to 50 epochs. The deeper CNN with Adam (~88%) outperforms the SGD variant (~87%) on convergence speed. The baseline (~84%) overfits visibly without regularization.

Conclusion. Pretrained features transfer effectively even across the ImageNet-to-MRI domain gap. Depth alone does not beat a well-regularized architecture. Glioma and meningioma remain the hardest class pair regardless of model, due to shared visual characteristics.

1. Introduction

Brain tumors affect many people each year. Early and accurate diagnosis improves treatment outcomes, but manual radiologist review is slow, expensive, and subject to inter-observer variation. Automated classification systems can support clinical workflows by prioritising high-risk scans and providing a second opinion.

CNNs have consistently outperformed classical machine learning on image classification benchmarks since AlexNet's ImageNet results in 2012. Later architectures — VGGNet, ResNet, Inception — pushed accuracy and sample efficiency higher still, extracting hierarchical features directly from pixel values without manual feature engineering.

Medical imaging poses specific challenges: labelled datasets are far smaller than natural image collections; class imbalance is common; and distinguishing glioma from meningioma requires fine-grained discrimination that even experienced radiologists find difficult. This project addresses three research questions: (1) how much do depth and regularization improve generalization over a bare baseline? (2) how does optimizer choice affect convergence? and (3) how much does ImageNet pretraining add over training from scratch on MRI data?

2. Dataset

The Brain Tumor MRI Dataset (Masoud Nickparvar, Kaggle) contains T1-weighted brain MRI images pre-divided into training and testing folders. Source: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

Class	Training Samples	Test Samples	Total
Glioma	~1,321	~300	~1,621
Meningioma	~1,339	~306	~1,645
No Tumor	~1,595	~405	~2,000
Pituitary	~1,457	~300	~1,757
Total	~5,712	~1,311	~7,023

Table 1: Dataset class distribution

Images vary in original resolution and aspect ratio — EDA revealed wide spread in width and height. All images are resized to 224×224 before training. Pixel values are normalised from [0, 255] to [0, 1] using a Rescaling layer. An 80/20 stratified file-level split separates training from validation. Data augmentation applied to training only: random horizontal flip, rotation $\leq 15^\circ$, zoom $\leq 10\%$, contrast adjustment, and translation $\leq 5\%$. The tf.data pipeline uses .cache() and .prefetch(AUTOTUNE) throughout. No-tumor is the most frequent class; glioma and meningioma are the least represented.

3. Methodology

3.1 Part A — Baseline CNN (From Scratch)

The baseline model follows the prescribed three-block structure. Each block is a Conv2D layer followed by MaxPooling2D, with filter counts $32 \rightarrow 64 \rightarrow 128$. After the third pooling layer the feature maps are flattened — a 224×224 input becomes a 100,352-element vector. Three Dense layers (512, 256, 128 units; ReLU; L2 regularization $\lambda=1e-4$) follow, with He normal initialisation throughout. A softmax output handles four-class prediction. Compiled with Adam ($\text{lr}=1e-3$) and categorical cross-entropy. No batch normalization or dropout — intentionally bare to establish a clear overfitting baseline.

3.2 Part A — Deeper CNN Architecture (build_cnn_v3)

The deeper model extends the baseline with a five-block VGG-style design, more than doubling the number of layers as required. Each block contains two stacked Conv2D layers, BatchNormalization, ReLU, MaxPooling2D, and SpatialDropout2D. Filter counts grow: $32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. SpatialDropout rates start at 0.10 in block 1 and rise to 0.25 in block 5. GlobalAveragePooling2D replaces the Flatten operation after block 5, producing a compact 512-element vector before the dense head ($512 \rightarrow 256$ units, BatchNormalization, Dropout 0.40 and 0.30). Label smoothing ($\epsilon=0.1$) is applied to cross-entropy loss. Class weights: glioma and meningioma at 1.5, no-tumor at 0.7, pituitary at 1.0.

3.3 Optimizer Comparison — SGD vs Adam

The same deeper architecture (build_cnn_v3) is trained separately with Adam and SGD+Nesterov to isolate optimizer as the single variable. Both use a 5-epoch learning rate warmup ($LR/5 \rightarrow$ full LR) to stabilise early training. ReduceLROnPlateau halves the rate after 3 flat validation epochs. EarlyStopping (patience=10) ends training and restores best weights.

Parameter	Baseline CNN	Deeper CNN (Adam)	Deeper CNN (SGD)
Optimizer	Adam	Adam	SGD + Nesterov
Learning Rate	1e-3	3e-4 (post warmup)	1e-2 (post warmup)
Momentum	—	—	0.9
Batch Size	32	32	32
Max Epochs	50	50	50
LR Warmup	None	5 epochs	5 epochs
Loss	Cross-entropy	CE + label smooth	CE + label smooth

Table 2: Hyperparameter comparison across models

3.4 Part B — ResNet50 Transfer Learning

ResNet50 (He et al., 2015) is loaded with ImageNet weights and the top classification layers removed. For 224×224 inputs, the base model outputs a $7 \times 7 \times 2048$ feature map. A custom head is added: GlobalAveragePooling2D $\rightarrow$ BatchNormalization $\rightarrow$ Dense(256, ReLU, L2) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(128, ReLU, L2) $\rightarrow$ Dropout(0.3) $\rightarrow$ softmax(4).

Phase 1 — Feature Extraction: All 175 backbone layers are frozen. Only the custom head trains for up to 10 epochs (Adam $lr=1e-3$, EarlyStopping patience=5). This picks up tumor-specific decision boundaries on top of ImageNet's generic visual features without disturbing them.

Phase 2 — Fine-Tuning: The last 30 ResNet50 layers are unfrozen. The full model trains for up to 10 more epochs at $lr=1e-5$ — $100\times$ lower than Phase 1 — preventing catastrophic forgetting while allowing the upper layers to adapt to MRI-specific patterns.

4. Experiments and Results

4.1 Baseline vs. Deeper Architecture

The baseline CNN shows a visible train/validation accuracy gap. Without batch normalization or dropout, the network memorises training patterns that do not transfer to unseen data. The five-block deeper model narrows this gap substantially through the combination of SpatialDropout (zeros entire feature map channels rather than individual neurons, preserving spatial structure), BatchNormalization (stabilises activations, adds implicit regularization), GlobalAveragePooling (slashes the parameter count before the dense head), and label smoothing (reduces overconfidence in predictions).

Model	Test Accuracy	Weighted F1	ROC-AUC	Epochs Run
Baseline CNN (Adam)	~0.84	~0.83	—	Early stopped
Deeper CNN (Adam)	~0.88	~0.87	—	Early stopped
Deeper CNN (SGD)	~0.87	~0.86	—	Early stopped
ResNet50 (Fine-Tuned)	~0.96+	~0.96+	~0.99	20 (10+10)

Table 3: Model performance on test set (highlighted row = best)

4.2 Computational Efficiency

All experiments ran on Kaggle's GPU environment (T4 or P100). `tf.config.list_physical_devices('GPU')` confirmed GPU availability before each run. The `tf.data` pipeline's `.cache()` and `.prefetch(AUTOTUNE)` kept the GPU fed between batches. The baseline is fastest per epoch (fewer parameters, no batch normalization) but overfits quickly. The deeper CNN runs slower per epoch due to extra conv layers and normalization steps. An `EpochTimingLogger` callback wrote per-epoch wall-clock times to a CSV for the SGD run. ResNet50 finishes in the least total elapsed time despite its depth — 20 epochs total versus up to 50 for the custom CNNs — because the frozen backbone Phase 1 runs gradient computation only through the custom head.

4.3 Optimizer Analysis — SGD vs Adam

Adam converges faster in early epochs. Adaptive per-parameter step sizes let it navigate the loss surface efficiently without precise LR tuning. SGD with Nesterov momentum requires the larger starting rate ($1e-2$) to traverse the parameter space without adaptivity, and needs the warmup period to stabilise early training. Both optimizers reach similar final accuracy on the deeper architecture, with Adam needing fewer epochs. For limited GPU budgets, Adam is more practical.

4.4 Ablation Study

To verify the contribution of regularization components, the deeper CNN was trained without SpatialDropout. The training/validation accuracy gap widened noticeably — the model began memorising spatial features in the feature maps, particularly for the meningioma class. BatchNormalization removal caused unstable loss curves and slower convergence. GlobalAveragePooling replacement with Flatten increased the parameter count sharply and raised the overfitting risk without accuracy benefit. These results confirm that the regularization combination is

load-bearing — removing any one component degrades generalization.

4.5 Challenges and Observations

Class imbalance: the no-tumor class has more samples and becomes the default prediction without correction — addressed with inverse-frequency weights (baseline) and custom per-class weights (deeper models). Glioma and meningioma are consistently the hardest pair — they share visual characteristics that both radiologists and models struggle to separate. SGD required higher LR and the Nesterov variant to converge within 50 epochs. Both Adam and SGD experiments include checkpoint resume via BackupAndRestore and ModelCheckpoint to survive notebook interruptions.

5. Transfer Learning with ResNet50

After Phase 1 (feature extraction), validation accuracy already matched or exceeded what the from-scratch custom CNNs achieved after far more epochs. This confirms that ImageNet features — edge detectors, texture filters, shape selectors — are general enough to transfer to MRI classification even across the domain gap.

Phase 2 fine-tuning pushes accuracy further. The frozen lower backbone layers retain low-level features (edges, textures, blobs) that appear in both natural images and MRI scans. The unfrozen upper 30 layers update to capture the specific patterns in tumor MRI data. The $lr=1e-5$ rate is critical — too large a rate would erase the useful backbone representations.

Transfer learning clearly outperforms training from scratch on this dataset. ResNet50 achieves ~96% accuracy in 20 epochs versus ~88% for the best from-scratch model in up to 50 epochs. The domain gap (natural images vs. MRI) is not an obstacle because low-level visual features are universal across image domains.

6. Conclusion and Future Work

6.1 Key Findings

Depth alone is insufficient — adding layers without regularization (baseline) leads to overfitting. The combination of BatchNormalization, SpatialDropout, GlobalAveragePooling, and label smoothing is what improves generalization, not depth itself. Adam converges faster than SGD on this dataset; both reach similar accuracy but Adam needs fewer epochs. Transfer learning from ImageNet works well even across the domain gap — ResNet50 beats all scratch models in a fraction of the training time. Meningioma and glioma remain the hardest classes regardless of architecture.

6.2 Trade-offs and Limitations

The baseline is interpretable and fast to train but generalises poorly. The deeper CNN is practical but ResNet50 surpasses it comfortably. ResNet50 is a black box — Grad-CAM visualisations would be needed for any clinical validation. The dataset (~7,000 images) suffices for an academic project but clinical deployment requires larger, more diverse collections from multiple hospitals and scanner types. The two-phase ResNet50 protocol runs only 20 epochs total; longer fine-tuning with a scheduled decay

could push accuracy higher.

6.3 Future Directions

Grad-CAM is the most useful near-term addition — it reveals which image regions drive predictions and would immediately flag if the model is responding to scanner artifacts rather than tumor tissue. EfficientNet or a Vision Transformer pretrained on RadImageNet would bring in more medically relevant prior knowledge than ImageNet alone. Ensembling Adam and SGD variants, or combining CNN and ResNet50 predictions, would smooth out individual model failures on edge cases. Pixel-level segmentation — producing tumor boundary maps — is the clinically relevant next step beyond classification. External validation on MRI data from other hospitals is the only reliable way to confirm the model learned general features rather than Kaggle-specific artifacts.